

ABSTRACT

WEATHER STATION

In this project, we will create a basic weather station using an Arduino board, a BME280 sensor, and an LCD shield. The BME280 is a high-performance environmental sensor capable of measuring temperature, humidity, and atmospheric pressure. It communicates with the Arduino via I2C protocol, making it easy to integrate and reliable for gathering real-time weather data. the weather station offers a practical application for home use, providing real-time information on local weather conditions. It can be further expanded by adding features like data logging or wireless connectivity for remote monitoring. By the end of the project, users will have a fully functional, low-cost weather station that is easy to build and customize for future improvements.

Through this project, users will gain hands-on experience in assembling electronic components, writing Arduino code, and troubleshooting common issues. The weather station provides accurate and real-time measurements of local environmental conditions, making it practical for home use or as a learning tool. this project can be further expanded by adding features such as data logging, wireless communication for remote monitoring, or even connecting to online weather platforms. By the end of the project, users will have a functional, cost-effective weather station that is both informative and customizable for future developments.

REFERENCE :

Arduino weather station research paper by research gate : [link](#) (online)

Arduino weather station research paper by MDPI :[link](#) (online)

